

Student Guide — primary / module-07-eleven-multiplication

IKS Curriculum

Table of Contents

Student Workbook

Student Workbook — $\times 11$ Multiplication — Ekādhikena Pūrveṇa (Primary)

This workbook is yours to write in. Use it for vocab, exercises, and reflection across all 10 days.

Vocabulary tracker

Fill in as we go:

- *Ekādhikena Pūrveṇa* — _____
- *sūtra* — _____
- *Anurūpyena* — _____

Day-by-day notes

Day 1 — The Hook — What Is the Trick?

What I learned:

One question I have:

Day 2 — The Carry Case

What I learned:

One question I have:

Day 3 — Why It Works — The Algebra

What I learned:

One question I have:

Day 4 — 3-Digit $\times$ 11

What I learned:

One question I have:

Day 5 — Speed Day

What I learned:

One question I have:

Day 6 — Anurūpyena — $\times 12$ to $\times 19$

What I learned:

One question I have:

Day 7 — Limits of the Method

What I learned:

One question I have:

Day 8 — Tirthaji and Historical Context

What I learned:

One question I have:

Day 9 — Mixed Practice

What I learned:

One question I have:

Day 10 — Final Assessment

What I learned:

One question I have:

Final reflection

After Day 10, write a paragraph: what is the most important thing you learned in this module?

Day 1 — The Hook — What Is the Trick?

Module: ×11 Multiplication — Ekādhikena Pūrveṇa (Primary) · **Time:** 30 min

Goal

Compute 23×11 , 35×11 mentally using the split-and-sum method.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

Today's idea (10 min)

- *Ekādhikena Pūrveṇa* (IAST: Ekādhikena Pūrveṇa; lit. “by one more than the previous” — sūtra 1 in Tīrthajī)

Introduce the concept gently using a story, a drawing, or a sensory example. The concept for today: **The Hook — What Is the Trick?**. Keep sentences short.

Activity (8 min)

A simplified version of *Mental Math Sprint Cards* (see activities/). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See homework.md.)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 2 — The Carry Case

Module: ×11 Multiplication — Ekādhikena Pūrveṇa (Primary) · **Time:** 30 min

Goal

Compute 87×11 , 65×11 correctly using the carry rule.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

Today's idea (10 min)

- *sūtra* (IAST: sūtra; lit. “aphorism / formula”)

Introduce the concept gently using a story, a drawing, or a sensory example. The concept for today: **The Carry Case**. Keep sentences short.

Activity (8 min)

A simplified version of *Mental Math Sprint Cards* (see activities/). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See homework.md.)

Teacher notes

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Day 3 — Why It Works — The Algebra

Module: ×11 Multiplication — Ekādhikena Pūrveṇa (Primary) · **Time:** 30 min

Goal

Derive $(10a+b) \times 11 = 100a + 10(a+b) + b$ and explain.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

Today's idea (10 min)

- *Anurūpyena* (IAST: Anurūpyena; lit. “proportionately” — sūtra used for $\times 12$, $\times 13$ extensions”)

Introduce the concept gently using a story, a drawing, or a sensory example. The concept for today: **Why It Works — The Algebra**. Keep sentences short.

Activity (8 min)

A simplified version of *Mental Math Sprint Cards* (see activities/). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See homework.md.)

Teacher notes

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- Keep new Sanskrit terms to one per lesson, repeated several times.
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Day 4 — 3-Digit $\times$ 11

Module: $\times 11$ Multiplication — Ekādhikena Pūrveṇa (Primary) · **Time:** 30 min

Goal

Compute $234 \times 11 = 2_5_7_4$ using the extended split method.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

Today's idea (10 min)

- *Ekādhikena Pūrveṇa* (IAST: Ekādhikena Pūrveṇa; lit. “by one more than the previous” — sūtra 1 in Tīrthajī)

Introduce the concept gently using a story, a drawing, or a sensory example. The concept for today: **3-Digit** $\times$ **11**. Keep sentences short.

Activity (8 min)

A simplified version of *Mental Math Sprint Cards* (see activities/). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See homework.md.)

Teacher notes

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- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 5 — Speed Day

Module: $\times 11$ Multiplication — Ekādhikena Pūrveṇa (Primary) · **Time:** 30 min

Goal

Complete a 50-problem mental-math sprint in 5 minutes.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

Today's idea (10 min)

- *sūtra* (IAST: sūtra; lit. “aphorism / formula”)

Introduce the concept gently using a story, a drawing, or a sensory example. The concept for today: **Speed Day**. Keep sentences short.

Activity (8 min)

A simplified version of *Mental Math Sprint Cards* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

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- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 6 — Anurūpyena — $\times 12$ to $\times 19$

Module: $\times 11$ Multiplication — Ekādhikena Pūrveṇa (Primary) · **Time:** 30 min

Goal

Extend the method to $\times 12$ and $\times 13$ using one-more-digit adjustments.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

Today's idea (10 min)

- *Anurūpyena* (IAST: Anurūpyena; lit. “proportionately” — sūtra used for $\times 12$, $\times 13$ extensions”)

Introduce the concept gently using a story, a drawing, or a sensory example. The concept for today: **Anurūpyena** — $\times 12$ to $\times 19$. Keep sentences short.

Activity (8 min)

A simplified version of *Mental Math Sprint Cards* (see activities/). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See homework.md.)

Teacher notes

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- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 7 — Limits of the Method

Module: $\times 11$ Multiplication — Ekādhikena Pūrveṇa (Primary) · **Time:** 30 min

Goal

Recognise when traditional long multiplication is faster than the sūtra method.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

Today's idea (10 min)

- *Ekādhikena Pūrveṇa* (IAST: Ekādhikena Pūrveṇa; lit. “by one more than the previous” — sūtra 1 in Tirthaji)

Introduce the concept gently using a story, a drawing, or a sensory example. The concept for today: **Limits of the Method**. Keep sentences short.

Activity (8 min)

A simplified version of *Mental Math Sprint Cards* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

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- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 8 — Tirthaji and Historical Context

Module: ×11 Multiplication — Ekādhikena Pūrveṇa (Primary) · **Time:** 30 min

Goal

Explain who Tirthaji was and why scholars debate the “Vedic” attribution.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

Today's idea (10 min)

- *sūtra* (IAST: sūtra; lit. “aphorism / formula”)

Introduce the concept gently using a story, a drawing, or a sensory example. The concept for today: **Tirthaji and Historical Context**. Keep sentences short.

Activity (8 min)

A simplified version of *Mental Math Sprint Cards* (see activities/). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See homework.md.)

Teacher notes

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- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 9 — Mixed Practice

Module: ×11 Multiplication — Ekādhikena Pūrveṇa (Primary) · **Time:** 30 min

Goal

Solve a mixed problem set using best-fit method per problem.

Materials

- Notebook, pencils, drawing paper

- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What’s one thing you remember from yesterday?*”

Today’s idea (10 min)

- *Anurūpyena* (IAST: Anurūpyena; lit. “proportionately” — sūtra used for $\times 12$, $\times 13$ extensions”)

Introduce the concept gently using a story, a drawing, or a sensory example. The concept for today: **Mixed Practice**. Keep sentences short.

Activity (8 min)

A simplified version of *Mental Math Sprint Cards* (see activities/). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See homework.md.)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 10 — Final Assessment

Module: $\times 11$ Multiplication — Ekādhikena Pūrveṇa (Primary) · **Time:** 30 min

Goal

Take the summative quiz and reflect on which method works best.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “What’s one thing you remember from yesterday?”

Today’s idea (10 min)

- *Ekādhikena Pūrveṇa* (IAST: Ekādhikena Pūrveṇa; lit. ““by one more than the previous” — sūtra 1 in Tirthajī”)

Introduce the concept gently using a story, a drawing, or a sensory example. The concept for today: **Final Assessment**. Keep sentences short.

Activity (8 min)

A simplified version of *Mental Math Sprint Cards* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

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- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Homework

Homework — ×11 Multiplication — Ekādhikena Pūrveṇa (Primary)

Light daily tasks. Each takes 5–10 minutes.

1. **Day 1 — The Hook — What Is the Trick?** · Draw about today’s concept. (~5 min)
2. **Day 2 — The Carry Case** · Draw about today’s concept. (~5 min)
3. **Day 3 — Why It Works — The Algebra** · Draw about today’s concept. (~5 min)
4. **Day 4 — 3-Digit × 11** · Draw about today’s concept. (~5 min)
5. **Day 5 — Speed Day** · Draw about today’s concept. (~5 min)

6. **Day 6 — Anurūpyena — ×12 to ×19** · Draw about today's concept. (~5 min)
7. **Day 7 — Limits of the Method** · Draw about today's concept. (~5 min)
8. **Day 8 — Tirthaji and Historical Context** · Draw about today's concept. (~5 min)
9. **Day 9 — Mixed Practice** · Draw about today's concept. (~5 min)
10. **Day 10 — Final Assessment** · Draw about today's concept. (~5 min)

Notes

- Homework is intended to consolidate what was learned in class, not introduce new content.
- If a child struggles, the parent should consult the teacher rather than push.
- Capstone project days (Module 15) may have lighter daily homework so students can focus on the project.